



Homeowner's Construction Guidebook





Homeowner's Construction Guidebook

Prepared By Homes and Communities

Published By CRS, Nepal

Illustration, Design and Printing WPS

Contents

Introduction to the Guide	1
Plot Assessment: Land Status and Ownership	2
Plot Assessment: Hazards	4
Plot Assessment: Planning	6
Budget: Grants	8
Budget: Loan Documentation Process	12
Budget: Assistance for vulnerable	14
Concept Design: Construction Typology	16
Concept Design: Cost Estimate	17
Design and Approval: Design Documents	19
Design and Approval: Building Permit System	21
Supplies and Services: Personnel	24
Supplies and Services: Construction Materials and Quality	25
Implementation: Construction Sequence	27
Implementation: Health and Safety	28
Implementation: Cash Flow Strategy	29
Implementation: Monitoring Workmanship	30
Annex: Financial Checklist	33
Annex: Technical Checklist	35

Introduction to the Guide

For households to develop plans and construct a house requires a number of processes. These include considering technical, financial, and social factors that combine to formulate a strategy for their construction. It can be a daunting task unless the households work in the construction industry or have constructed previously. There will be variations in the construction process and timeline but these center around the same key steps like assessing plots, managing finance, selecting consultants and contractors, managing construction, adhering to municipal compliance, etc.

This guide provides a general overview of construction stages and how to plan each stage. The overall construction process has been divided into six stages:

1. Plot Assessment
2. Budget
3. Concept Design
4. Design and Approval
5. Supplies and Services
6. Implementation

Each of these stages have been subdivided into many chapters. Let's investigate each of the stages and their sub-chapters in further detail.



Plot Assessment: Land Status and Ownership

How do I assess my plot?

Welcome to the first step to owning a house! The first step of course is selection or purchase of suitable plots for construction. You will need to go through three different stages to understand the various criterias that render the plot's suitability to house constructions:

- Land status and ownership
- Common hazards
- Land use planning and zoning

Let's move to the first stage "Land status and ownership".

What do I need to prove land ownership?

Land is categorized into different categories by the Ministry of Land Management, Cooperatives and Poverty. Some common land types are:

- Private land: It is owned by an individual or group of individuals. It can be acquired by purchase, inheritance, or government allocations. It is subject to yearly taxation.
- Public/Government land- It is national territory that includes public land and government land and no private constructions are allowed here.
- Guthi land – It belongs to religious bodies and comprises temples and monasteries as well as religious schools and hospitals. Guthi land can not have private ownership.



To construct a private house you will need to identify a plot where you can prove your ownership. Private Land Ownership Registration Certificate or lalpurja proves your ownership.

Private Land Ownership Registration Certificate (Lalpurja) is a document registered at the Survey Department and Land Revenue Office of the Ministry of Land Management, Cooperatives and Poverty. The certificate clearly shows details about:

- Registered owner's details
- Land plot/parcel number
- Land area
- Land type

Please ensure all these details are represented in your Private Land Ownership Registration Certificate (Lalpurja). If you already own a plot and have the certificate, you can move ahead to the next stage of assessment. If you are yet to purchase a plot, the following stage will help you understand the process to purchase a plot.

How do I purchase a plot?

You will first need to select a private plot that meets your requirements and budget through your known connections or real estate company. To issue the Private Land Ownership Registration Certificate (Lalpurja) under your name, you will need to visit the Survey Department with following documents:

- Private Land Ownership Registration Certificate (Lalpurja) of the seller (from whom you are to purchase the plot)
- Citizenship of both buyer and seller
- Land tax clearance receipt of ward office
- Ward recommendation on land verification by the adjacent neighbors of land (Char Killa)

Understood the process? Wait for the purchase as you need to go through two other stages prior to finalizing the plot.

Plot Assessment: Hazards

Is my plot suitable for construction?

The second stage now is to assess the plot for safety from different hazards prior to finalizing or purchasing the plot. Hazard is a natural or man-made event or phenomenon that may cause loss of life, injury or other health impacts and property damage.

A very general checklist on hazards will help you to assess your land on your own. You can always consult an experienced engineer to assess your plot from the common hazards. With the details below, check your plot against the common hazards.

Can you please assess my plot from hazards?



Sure, let's go through a checklist to assess the plot from hazards.



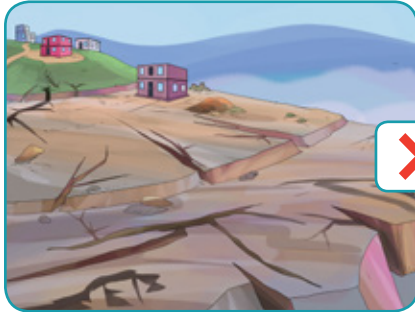
High ground slope higher than 20 degrees



Flood prone area with no setbacks as guided by municipal by-laws



Stone falling areas



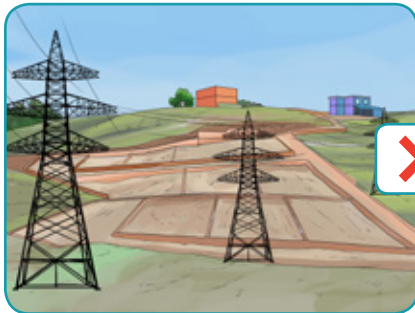
Seismic ground faults with visible fault lines of earthquake



Refilled plots with loose sand/silts



Marshy and waterlogged area



Electric transmission line with no setbacks as guided by municipal by-laws



Large tree roots area

Done with the assessment? If your plot is safe from the above hazards, you can move to the next stage to owning your house.

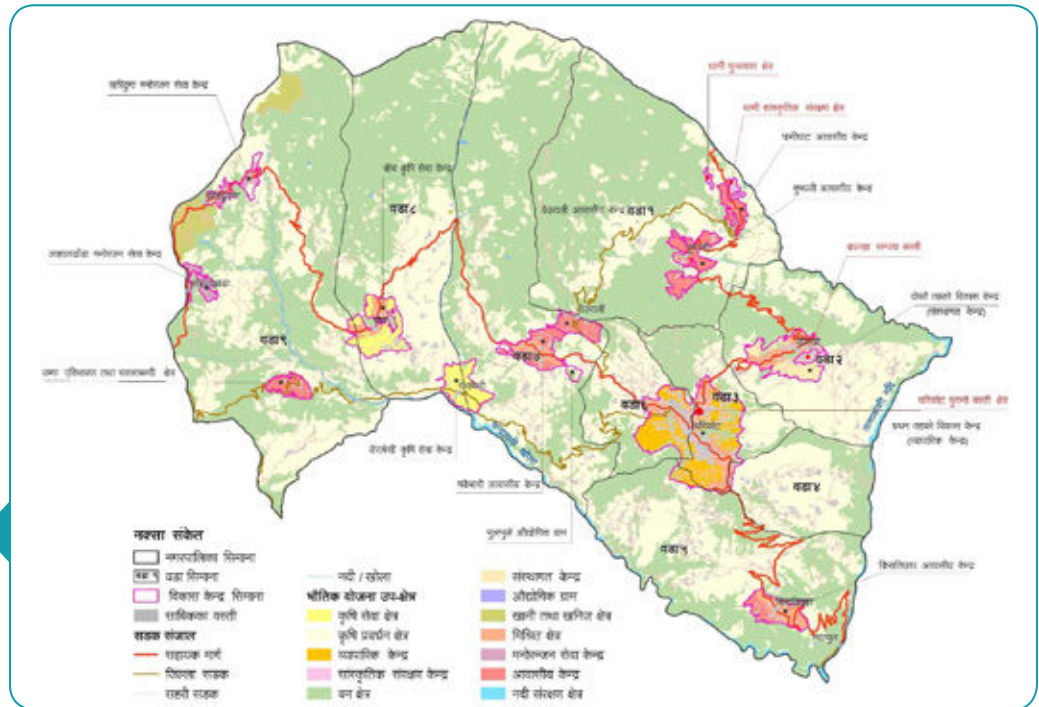
Plot Assessment: Planning

Am I allowed to build on my plot?

Once you have assessed the safety of your plot from various hazards and confirmed its suitability for construction, you will now need to understand the land use policies and zoning regulations of the municipality where your plot is located. It is essential to understand the land use and zoning policies of the municipality as it regulates where and how you can construct.

Land use planning refers to the process by which land is planned in such a way so as to promote more desirable social and environmental outcomes, orderly development of land as well as a more efficient use of resources. Generally, land use plans have built up (residential) areas and non-built up (agricultural, forest, water bodies, conservation, green spaces) areas. Zoning, whereas is a planning control tool of the municipalities for regulating the built environment and its functionality.

The image here is a land use plan of Bhimeshwar Municipality, Dolakha prepared by NSET. It shows the land use and zonings like Agriculture zone, Agriculture promotion zone, Commercial zone, Cultural conservation zone, Forest zone, Institutional zone, Mines and Minerals zone, Mixed use zone, Recreational zone, Residential zone and River conservation zone



Zoning divides the built-up areas into various zones like residential, dense mixed residential, urban expansion, core areas and old settlements, commercial, industrial, institutional, etc. Zones are specific to the municipalities and can differ from other municipalities. The easy way to list out the land use and zoning regulations appropriate to your plot is referring to the municipal by-laws. Municipal by-laws are ready-to-use guidance applicable within the municipality. It consists of minimum parameters that need to be followed for constructions adhering to the land use and zoning policies of the municipality. You can obtain the by-laws from the municipality, or an experienced engineer can guide you on identifying the minimum requirements you will need to follow to adhere to the land use and zoning policies of the municipality.

Municipal By-laws



Plot size

Minimum area of plot



Height

Height of building or construction from the average floor level to the top/highest surface level with respect to the flat roof structure.



Floor Area Ratio

FAR is the ratio of total building floor area to the area of its zoning lot. For example, on a 10,000 square foot plot in a municipality area with a maximum FAR of 2.0, the floor area cannot exceed 20,000 square feet,



Ground Coverage

Ground Coverage indicates how many percent the total ground/plot are permitted to be covered by buildings including accessory buildings. For example, on a 10,000 square foot plot in a municipality area with a maximum of 60% Ground Coverage, the building plinth area cannot exceed 6,000 square foot.



Density

Number of buildings, dwellings, households, people or the amount of floor space per hectare.



Floor Area

Floor Area is calculated on the basis of the external dimensions of buildings including all storeys, except floors entirely located below ground level, of any building. In calculating the floor area, as well account shall be taken of secondary structures as balconies, attics, recessed balconies or patios, enclosed porches and floor area devoted to accessory use. Any area constructed and used for vehicle parking or loading of vehicles shall not be included as floor area.



Right-of-way

A land corridor designated or constructed for the use of public access, vehicular traffic circulation and the location of public utilities, such as pathways, easements, roads and highways, regardless of the ownership of the land and utilities.

Now that you have completed and come through these assessments, you can finalize or purchase your plot to own your house!

Budget: Grants

How can I finance the construction?

The second stage is planning to finance your construction. Yes, you yourself are responsible for managing finances for construction. Planning and managing finance is a daunting task. But if you understand the different sources of how you can finance your budget, you can prepare your financial management strategy.

There are some provisions by the government to support financing for housing construction. It is essential to know these provisions to plan the financial management. Municipalities can provide you with the information. So, prior planning the financial management, you need to be prompt and visit the municipality office to get the information.

Thank you for coming by to get the information on different housing finance support by the government. Please be mindful that these provisions might change and there can be new provisions. You need to keep updating yourself by visiting the municipality before and during the construction. For now there are two provisions for housing finance:

- If you are living in heritage area, municipality provides Heritage Grant of 50,000
- If you are the victim of multi-hazards, NDRRMA provides you different grants like:
 - » Reconstruction Grant for complete housing reconstruction of 300,000 in terai, 400,000 in hills and 500,000 in mountains
 - » Retrofit Grant for partial strengthening of house of 50,000
 - » Land purchase for resettlement of 300,000



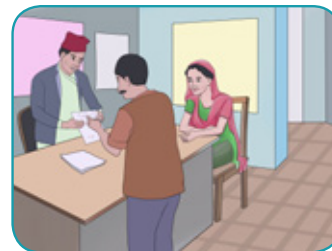
As per the National Disaster Risk Reduction and Management Authority (NDRRMA) “Monsoon Disaster Affected Private Housing Reconstruction and Rehabilitation Grant Procedure 2077” and “Fire Disaster Affected Private Housing Reconstruction and Rehabilitation Grant Distribution Procedure 2078” for you to get this facility, your house must be damaged by flood, landslide or fire for. The following process and grants have been provided for private housing reconstruction.



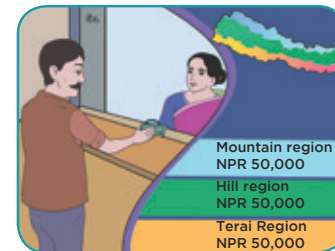
1. Technical inspection of the damage by municipal technical persons



2. Beneficiary List publication at municipality after the verification process



3. Partnership Agreement with the municipality



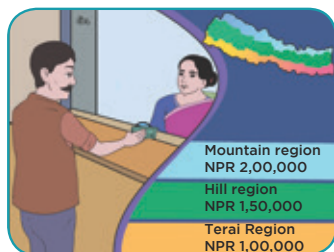
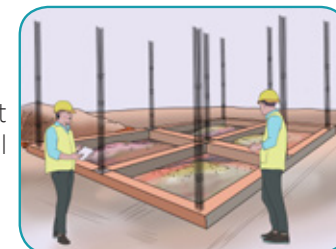
4. First tranche release in beneficiary's bank account

Mountain region	NPR 50,000
Hill region	NPR 50,000
Teral Region	NPR 50,000



10. Completion certificate issued by municipality after complete construction

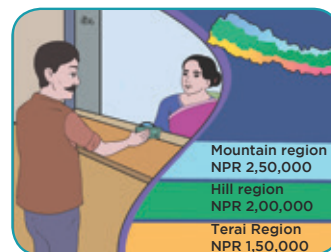
5. Municipal building permit application and approval



9. Third tranche release in beneficiary's bank account



8. Construction till roof level and technical inspection by municipal technical persons



7. Second tranche release in beneficiary's bank account

Mountain region	NPR 2,50,000
Hill region	NPR 2,00,000
Teral Region	NPR 1,50,000



6. Construction till DPC level and technical inspection by municipal technical persons

Budget: Loans

How can I finance the construction?

It is obvious that you have a limited budget to construct your house. Availability of the whole budget at once is not always possible, so you need to look for options to source your budget. The next finance option you have is Bank Loans. Loan is the method of lending money by one or more individuals, organizations, or other entities to other individuals, organizations, etc. The recipient incurs a debt and is usually liable to pay interest on that debt until it is repaid as well as to repay the principal amount borrowed.

Bank manager officer says:
Namaste! There are different types of loan you can request at our bank.



Namaste, I am planning to construct a house, can you share about different loans that I can get?

1. Home Loan

A home loan is an amount of money that an individual borrows from a bank at a certain rate of interest to be paid with the EMI every month. The property/house is taken as a security by the banks for the Home Loan.

2. Personal Loan

A personal loan is an amount of money you can borrow to use for a variety of purposes. You may use a personal loan to consolidate debt or pay for home renovations.

3. Business Loan

A business loan is a loan specifically intended for business purposes.

4. Concessional Loan

Concessional loan is a loan that provides for different purposes such as agriculture production, business, house construction, skill development with business, etc. In this loan, the government provides a subsidy in interest rate to clients.

The common terminologies we need to understand for accessing bank loan are:

- **Interest Rate:** Interest rate depends on the different types of loan. Interest rate will be different as per the base rate of banks. Base rate changes every three months and banks publish base rate and interest rate as per the rules of Nepal Rastra Bank.

- **Loan Tenure:** The maximum flexible loan tenure period is up to 35 years.
- **Equated monthly Installment (EMI):** EMI is a pre-agreed monthly payment to be paid by the borrower to the creditor on a fixed day of each month. EMIs include both interest and principal (amount borrowed) of each month.
- **Collateral:** It is something, usually property (house/land) that you pledge as security for the repayment of a bank loan. If you fail to repay the loan, collateral is forfeited by the bank.
- **Bank Account:** An account with a bank created by the deposit of money or its equivalent and subject to withdrawal of money (as by cheque or passbook).

To open a bank account, following documents are required:

- **Copy of citizenship:** It is mandatory to have citizenship to open a bank account. If you are taking a concessional loan, you need to submit citizenship of all your family members.
- **Passport size photograph:** You need to have four copies of your (account holders) passport size photographs.
- **Copy of bills of water or electricity:** If your permanent address (district) is different from the bank address, you will need to submit a copy of either water or electricity supply of your temporary address.
- **Bank application form:** You will need to fill an application form at the bank, the bank staff will help you fill out the form if help is required.

Budget: Loan Documentation Process

What are the documents required for bank loans?

Now that you have understood the different types of bank loans and opened your bank account, let's learn about the bank loan documentation process. For the bank loan application, you will need to submit different documents at different stages.

“Namaste! Sure I will. You will first need to understand the three different stages of the loan documentation process and then I will provide you with a check-list of documents you will require for each stage.”



“Namaste, I am planning to construct a house. I have already opened a bank account. Now I need to apply for a loan to construct my house. Can you share in detail about the loan documentation process?”

The three stages of loan documentation process and the documents required are:

- **Applicant Identification Process:** To introduce/identify yourself as the applicant to the bank, you will need to submit personal documents that introduce you which are:
 - » Passport size photograph (2 copies)
 - » Citizenship/Passport
 - » Bank account statement (deposit & loan) of at least 6 months
 - » Location map (residence/office)
 - » Based on relevance, you may also require,
 - » Relationship certificate
 - » Marriage certificate
 - » Firm/Company registration certificate
 - » PAN/VAT certificate
 - » Article of Association (AOA) and Memorandum of Association (MOA)
 - » Partnership deed
- **Collateral Declaration Process:** There are risks that the clients discontinue to pay off their EMIs. Therefore to prevent the investment from sinking, banks have made it compulsory to submit collateral documents equal to or more than the amount of loan requested. The documents are:
 - » Land ownership certificate (Lalpurja)
 - » Rajinama or Bakaspatra or Ansha-banda
 - » Survey department/Malpot Receipt
 - » Tax clearance of building
 - » Blueprint and trace map of land
 - » Verification by four neighbors of land/Char Killa Pramanit
 - » Municipality building permit
 - » Building completion certificate/Nirman sampanna patra
- **Income Source Declaration Process:** You will need to prove your monthly and annual income source to the bank. If the income source is strong, the loan disbursement process can be faster. The documents required are:
 - » Salary certificate
 - » Rental agreement (If any) along with tenant's identification
 - » Pension Certificate (Pension Patta) (if any)
 - » Audited financials of last 2 fiscal years (If business income)
 - » Copy of renewed/valid blue book of vehicle along with route permit (If income is from commercial vehicle)
 - » Salary certificate, work permit, valid passport with visa, pay slip and bank account statement (If income is from foreign employment)
 - » Latest tax paid receipt or tax clearance certificate for all income (wherever applicable)

Budget: Assistance for vulnerable

What are the alternative sources to finance construction for the vulnerable?

There are various provisions of housing assistance for the vulnerable from the government side as well. If you have been identified as vulnerable by your municipality/rural municipality, you can consult with your ward/municipality to understand and access available housing assistance schemes.

A. Government housing schemes

There are some housing programs of the Nepal government for the vulnerable which are implemented by the province government. “Janata Aawas” is one such housing program where the government provides NPRs. 350,000 in four tranches for construction of a house. Municipalities identify the vulnerable households based on the criteria set by the provincial/national government. You can consult the municipality office to know about the housing assistance programs.

B. Cost subsidy of construction materials

Municipalities can sometimes facilitate the process for the cost subsidy of construction materials to assist the vulnerable. Municipalities coordinate with the locally identified construction material suppliers to provide a cost subsidy of 7-15% for the vulnerable. For this process, a recommendation letter from the ward/municipality office is required.

C. Deprived sector investment through commercial banks

Commercial banks and microfinances can provide housing loans to vulnerable households from the deprived sector investment that has been provisioned by the Nepal Rastra Bank (NRB). The optimum loan amount depends on the internal policy of banks and microfinances. For instance, Nabil bank provides Amount as deprived sector loan.

You can coordinate with the municipality to identify such loan schemes and apply for the loan. Following documents are required for the deprived sector investment concessional loan:

- Participation Agreement Card with government (If any)
- Recommendation letter from municipality
- Recommendation letter from ward office to municipality (If necessary)
- Personal legal documents (Citizenship of all family members, land certificate, income source-if you have)

D. Revolving fund

Revolving fund is a practice started by municipalities to support the construction of vulnerable households of the Nepal earthquake of 2015. A small amount of revolving fund is set up by the municipality based on the number of vulnerable households. The households take the fund and after they receive the reconstruction tranche, they return the revolving fund. This helped the households to construct quickly without waiting for the government tranche. Municipalities can link this concept with livelihood options to support vulnerable households to construct their house.

E. Local government provisions

Municipalities can link the housing assistance for the vulnerable households with its various programs and can also link the support with various development organizations working in the municipality under the same modality. According to the Housing Act- 2018, article 14, local government/municipalities can provide shelter for the landless and vulnerable households. Based on this act, municipalities and development organizations can provide housing support with 50/50 cost contribution.

The Prime Minister Self Employment Programme (PMSEP) is one of the programs being implemented by all municipalities and as per the guideline, municipalities can mobilize manpower for the construction of houses of the vulnerable.

F. Home Insurance

Home Insurance policy is designed to cover the damages related to properties from the private insurance company. This type of policy covers damages from various risk factors such as flood, earthquakes, landslides, storms, explosions, fire, riots, and more. Home insurance ensures safety for your house and protects you from any acts that may be out of your control. If you have a home insurance policy, at your vulnerable state, the insurance helps you to cover the damages of your house.

The following documents are required to apply for the insurance payback:

- Copy of citizenship
- Copy of land certificate
- Self declaration of building's total value
- Application form provided by insurance company
- Home insurance policy



Concept Design: Construction Typology

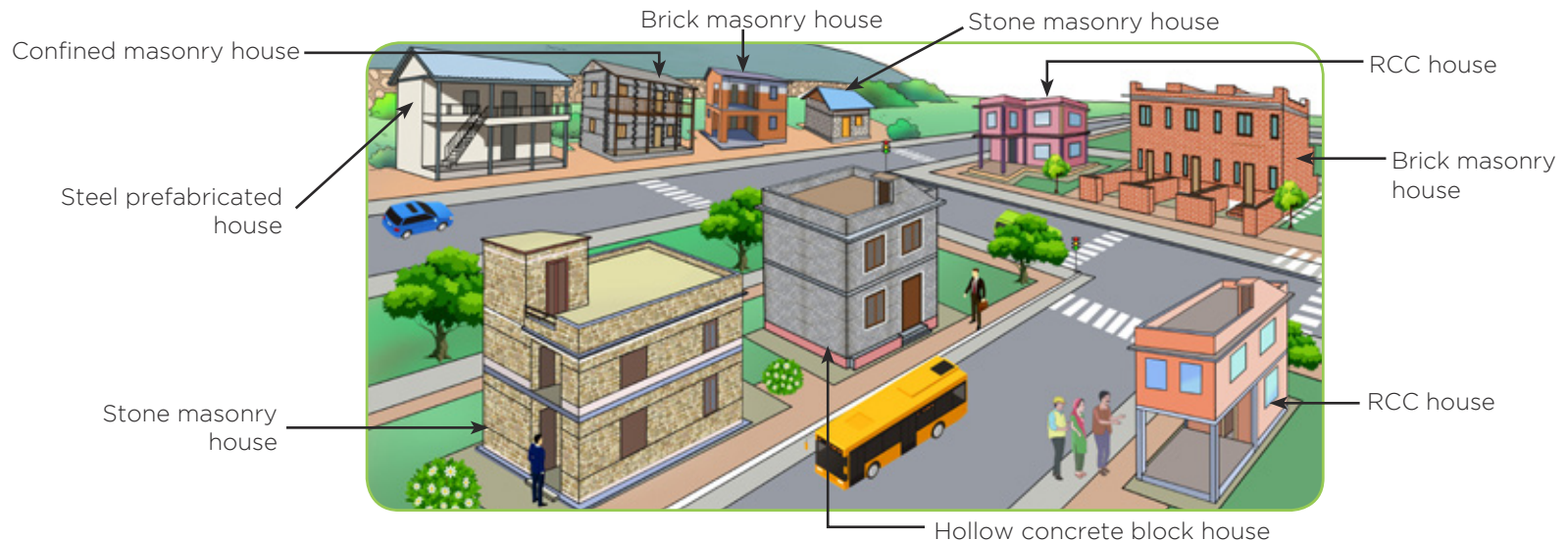
What can I build with my budget?

Once you have a tentative idea of your budget, you can start on the concepts to design your house. The first stage obviously is to identify the type of house you want to construct. Based on your budget, you can choose the three major components of your house:

Structural component: It is the component which bears the load of your house. You can choose Load bearing masonries of bricks, stones, hollow concrete blocks or RCC structure or Steel structures as the structural component of your house.

Infill component: It is the component which walls/clads your house between the structural components. You can choose bricks, stones, hollow concrete blocks, prefabricated materials, etc. as infill components. Load bearing structural components behave as a whole unit, i.e., do not have infill components.

Floor: It is the component which provides the horizontal base for each storey. You can choose Flexible floors like wood, bamboo, steel, etc. or Rigid floors like RCC slab.



Concept Design: Cost Estimate

What can I build with my budget?

Yes, it is difficult to choose from the different construction typologies unless you have an idea of the tentative cost of construction. The best way to estimate the tentative cost of your house is to consult with your family and friends who just recently constructed a house. Engineers and contractors are the best persons to get a preliminary estimate. They all will provide you with an average cost per square meters/feet based on which you can have the preliminary estimate.

For example, **if you are planning to construct a RCC house 30 feet X 30 feet house with two and a half storeys:**

Area of house: $30 \times 30 \times 2 + 15 \times 15 = 2,025$ sq.ft.
 Rate of 1 sq.ft. provided by contractor (per your location) = NPRs. 2,700
 Total estimate of your house = Area X Rate (per sq.ft.) = $2,025 \times 2,700 =$ NPRs. 5,467,500

This is a very tentative estimate, and it depends on the location as well. So, you need to be mindful of obtaining the tentative up-to-date rates specific to your location.

For a general comparison of budgets according to different housing typologies, you can also refer to the table here. The estimate is for major cities with an easily accessible marketplace.

Comparison:

Cost Estimate and Comparison between different typology						
		Different typologies maintaining traditional outlook				Brick Masonry in Cement with wooden floors
SN	Particulars	Stone Masonry in Cement	Brick Masonry in Cement	Confined Masonry	Reinforced concrete	(Traditional)
1	Total Area in sq.ft.	750	1900	1800	1800	1800
2	No. of Story	2	2+ Attic	3+attic	2+attic	2+attic
3	Rate of civil work per sq. ft.	1450	2100	3150	3400	4750
4	Total cost in NPR	1,087,500	3,990,000	5,670,000	6,120,000	8,550,000
5	Total Cost in NPR/1000 sq.ft.	1,450,000	2,100,000	3,150,000	3,400,000	4,750,000
Cost of construction of a single storey (in NPR)						
6	Small sized (600 sq.ft)	870,000	1,260,000	1,890,000	2,040,000	2,850,000
7	Medium Sized (850 sq.ft)	1,232,500	1,785,000	2,677,500	2,890,000	4,037,500
8	Large sized (950 sq.ft)	1,377,500	1,995,000	2,992,500	3,230,000	4,512,500

- If you want to construct a two storey Stone masonry in cement house with a total area of 750 sq.ft.

(combined area of both storeys), the cost of one sq.ft. is NPRs. 1,450 whereas the total cost of construction is NPRs. 1,087,500.

- If you want to construct a two and a half storey RCC house, the rate up to a maximum of 1,800 sq.ft. is NPRs. 3,400. Similarly, if you want to construct a Brick masonry in cement house, the rate up to a maximum of 1,800 sq.ft. is NPRs. 2,100. So, for similar house size, RCC costs NPRs. 1,300 more than brick masonry.
- If you want to construct only a single storey house, the unit rate is different compared to two storeys. This is due to the high cost of foundation. So, if you want to construct a traditional brick masonry in cement house, the total cost for an area up to 600 sq.ft. is NPRs. 2,850,000, for an area exceeding 850 sq.ft. up to 850 sq.ft. is NPRs. 4,037,500 and area exceeding 850 sq.ft. up to 950 sq.ft. is NPRs. 4,512,500.
- If you want to construct a confined masonry house of three storeys with a total area of 1,800 sq.ft., the unit rate is NPRs. 3,150 and the total cost is NPRs. 5,670,000. Similarly, if you want to construct a single storey confined masonry house of 600 sq.ft., the total cost is NPRs. 2,040,000, for an area exceeding 850 sq.ft. up to 850 sq.ft. is NPRs. 2,890,000 and area exceeding 850 sq.ft. up to 950 sq.ft. is NPRs. 3,230,000.

This is an example of how you can obtain such tentative comparison rates from experienced engineers or contractors. The parameters you will require are:

- total number of storeys
- total area
- construction typology and
- unit rate.

This will help you to identify which typology of house and how many storeys you can construct within your budget.

There also are some tools and softwares available online for the estimate. One example is the **House Cost Calculator**. It is a simple web and mobile based application developed by Housing Recovery and Reconstruction Platform (HRRP) with support of CRS and NSET. It rapidly calculates the cost of house construction for various housing typologies of different shapes and sizes fed in the tool. It can be accessed through HRRP website. You are asked to input the construction typology, location and the tentative dimensions of the house. Based on the pre-registered data of construction materials cost, labor cost and transportation cost, you will be able to get the tentative cost of your house.

Design and Approval: Design Documents

What is the design process?

You have your conceptual design ready? Yes, then you can proceed to the final design phase. You can identify a design consultant who can be a civil engineer or an architect or a registered engineering/architectural firm, through your contacts, internet sourcing or from the municipality. Design consultant will help you to materialize your concept into final designs adhering to the standard bylaws and building codes of the municipality.

We have talked in detail about the municipal bylaws in the “Plot Assessment: Planning” section earlier. The bylaws set a guiding principle for constructions based on the land use and zoning regulations of the municipality which needs to be considered for the final designs. Bylaws generally guide the overall settlement planning.

Building codes, whereas refers to the national standards which guides the houses/buildings to achieve seismic safety by setting out the minimum requirements for earthquake resistant constructions. There are four types of building codes which need to be followed based on the purpose of your construction. For house

constructions, you will need to follow the relevant one from two codes:

Professionally Engineered Buildings: If you are planning to construct a house with the plinth area (building area at ground floor) more than 1,000 sq. ft., or more than 3 storeys or span (distance between pillars) more than 4.5 m or irregular shapes, the house must be designed and constructed under the supervision of engineers.

Mandatory Rules of Thumb: If you are planning to construct a house with the plinth area less than 1,000 sq. ft. and less than 3 storeys and span less than 4.5m and regular shapes, the house can be constructed by technicians. This is applicable to the areas where no professional engineers service is available. Generally, all municipalities have a mandatory provision of building permit and drawings prepared by professional engineers are required. In case there is no provision of building permit and compliance to the building codes are sufficient for the municipality, mandatory rules of thumb can be followed.

We will help you to prepare the designs and drawings based on your concept and integrate all the codes and bylaws. The first step now is the finalization of designs which we will do by consulting you. We will then prepare the final drawings. Two different drawings are required:

- Architectural drawings
- Structural drawings



We have worked on our concept and budget for our house. How should we move ahead with the design process?

Architectural drawings consist of the layout of the rooms which we call plans and the layout of the four walls which we call elevations. Structural drawing, whereas consists of the layout and details of beams, pillars, and all structural members of the house. These are the mandatory documents which will guide you in the field implementation or construction of your house. The contractors or engineers you hire for construction will easily understand these drawings. Furthermore, we can help to prepare the 3D pictures of the design of your house.

You will need to submit these drawings, both Architectural and Structural to the municipality for the building approval. If you are planning a professionally engineered building, we will also need to prepare a Structural Analysis Report which consists of the analysis of the structural components of the house. This report also needs to be submitted to the municipality for approval.

Apart from these drawings, you will require Bill of Quantities (BoQ) and Detailed Estimate for the construction implementation. BoQ provides you the quantities of construction materials and labor for the complete construction of your house and Detailed Estimate provides you the quantities and cost of construction materials and labor for each stage of construction. We can help you prepare these documents. You can also get the tentative quantities from the contractor you will assign for your construction but that will be a lump-sum one. If you are planning to lead and supervise your construction, these documents can help you plan and budget all stages of construction.

After the preparation of all these drawings, the next stage now is the Building Permit Approval from the municipality.

Design and Approval: Building Permit System

What is the approval process?

Once you have your designs and drawings completed, you are ready to move forward to the approval process. Yes, you need to get approval from the municipality prior to constructing your house and that approval process is called Building Permit System. It is a regulatory process to register and approve the designs and construction within the municipality. At first, you will need to obtain a Building Permit Book from the municipality. The engineer/architect who helped you prepare the drawings will help you to fill the details required in the book. The approval is a long process which ends only after you receive the Final Completion Certificate from the municipality. At various stages, you will need to submit required documents at the ward and municipality. The process below will help you prepare and move ahead with the building permit system.



1. Preparation of house drawings



2. Obtaining building permit book from the municipality



3. Registering drawings and building permit book at the municipality



4. Drawings and details on building permit check by technical persons at municipality



5. 15 days notice by ward for neighboring plot owners



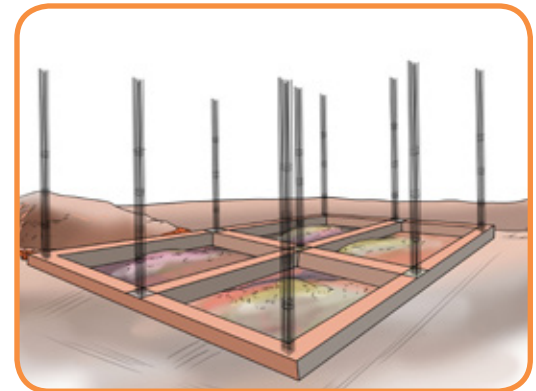
6. Field investigation by municipality field surveyors



7. Tax payment at the ward office



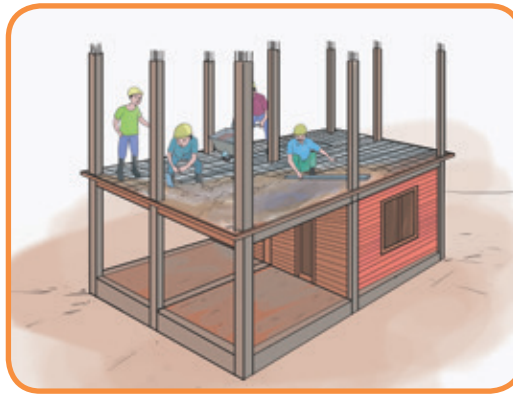
8. Temporary construction permit till plinth level issued by the municipality



9. Construction till plinth level



10. Second permit issued by the municipality for the construction of superstructure



11. Construction of superstructure



12. Building completion certificate issued by the municipality

Supplies and Services: Personnel

How can I get the best value when constructing my house?

Applied for the building permit at the municipality? It will take some time, so you can utilize your time to understand and plan for the supplies and services required for the construction. This will help you effectively manage and budget for various stages of construction. Let's begin with the personnel you will need to engage for the construction.

Contractors: A contractor is the one who manages the overall construction materials and workforce to complete the house. Instead of contractors, you yourself can manage the overall construction process. Contractors are not associated with municipal registration, so you can hire a contractor through recommendations from friends, families, relatives, and neighbors but it is essential to confirm the previous experiences of the contractors, the types of skilled and unskilled workers engaged, etc. before handing over the construction to the contractor. It is advisable to carry out written contract with the contractor with as much details as possible to avoid conflict in the future.

Labor: These are the skilled and unskilled manpower directly engaged in the construction of the house. Generally, the different types of construction workforce are:

- **Skilled Labor:** Most of the municipalities these days have made it mandatory to engage skilled masons in construction. Skilled masons are the certified masons who have enrolled in 7-days and 50-days "Mason Training Program". Engaging skilled masons helps to achieve the quality of construction and to meet all standards of construction specified by the municipality.
- **Unskilled Labor:** Unskilled manpower are the laborers and supporters who assist the skilled manpower during construction.

Suppliers: Suppliers are the local vendors who provide the construction materials. Based on the quality, stock, and delivery time, you can select single or multiple suppliers. Instead of dealing with the suppliers, you can also hand over the responsibility of identifying and dealing with the suppliers for timely delivery of construction materials to the contractor.

Supplies and Services: Construction Materials and Quality

How can I get the best value when constructing my house?

To plan and confirm the suppliers, you will first need to understand the construction materials required and their quality. The construction materials will depend upon the construction typology you choose. If you choose an RCC building with brick walls, you will require: Bricks, Cement, Sand, Aggregate, Steel bars, Wood, and Water as the primary construction materials. You will also require other materials such as: Paints, Floor finishes (Tiles, Parqueting, Marbles), UPVC/Aluminum, etc. based on the details and finishes you choose. Here we talk about the basic requirements of the primary construction materials that are applicable to almost all construction typologies.



Brick

- Should be well burnt and have uniform color
- Should have regular edges and without cracks
- Should produce a metallic sound when two bricks are struck
- Should not absorb more than 25% water to its weight after dipping in water for 24 hours
- Should not break into multiple pieces when dropped from 3 feet height



Cement

- Should be Nepal Standard certified, it will be mentioned in the bag
- Should be used within 3 months from the date of manufacture.
- Should be warm when hand is inserted inside the bag
- Should be in an air-tight bag
- Bag should not be punctured with a hook
- Should be warehoused in dry, airtight, and watertight store
- Should not form lumps



Aggregate

- Should be well graded i.e., consist equal proportion of different sizes of aggregate
- Should be hard, clean, and tough
- Should be free from organic matter, oily substances, and other contaminants
- Maximum size should be less than or equal to 40mm for concreting columns and beams
- Maximum size should be less than or equal to 20mm for concreting thin slabs



Stone

- Should not be easily broken, soft or light
- Should not have any deep cracks
- Should have regular edges with at least 2-inch thickness and 6-inch length
- Should not be round



Sand

- Should be clean, granular, and rough
- Can be river or quarry sand but should not have much moisture
- Should not be contaminated with organic, oily, or other substances



Steel bars

- Should not have rust
- Should not break when bent
- Should not have any cracks
- Should have weight and quality clearly mentioned in the product card



Wood

- Should be well-seasoned
- Should not have any splits, or holes, or knots



Hollow Concrete Block

- Should be uniform and free from cracks and broken edges
- Should be free from honeycombing i.e., large voids inside concrete
- Should not absorb water more than 10% of its weight



Water

- Should be clean and potable i.e., drink worthy

Based on your availability, preference, budget, and construction knowledge, you can choose the construction personnel. You can manage the whole construction on your own including both personnel and materials. You can hire a contractor to manage the personnel and look after procuring and managing the materials yourself or you can assign the contractor with both personnel and materials management. The checklist above will help you to identify the right quality of construction materials.

Implementation: Construction Sequence

What are the basics I need to understand for the implementation of my house construction?

Once you have identified the personnels for construction, suppliers to supply quality materials and have obtained the temporary building permit from the municipality, you can now start the implementation, i.e. construction of your house. You will need to first understand the basics of the construction process. All type of house construction have a similar construction sequence:

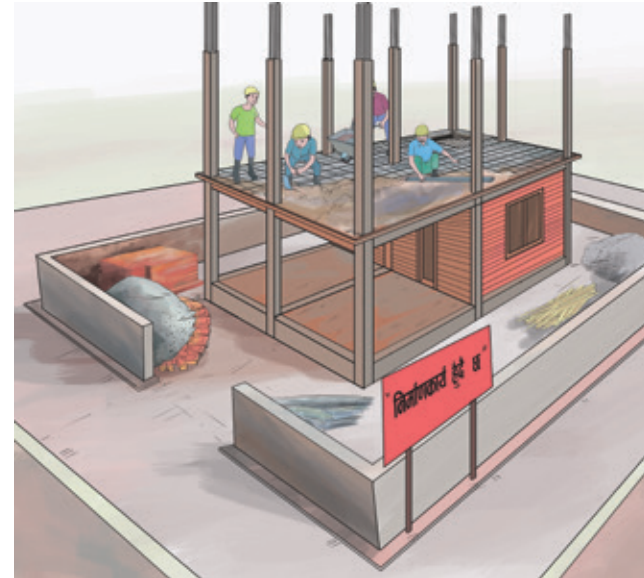
1. **Prepare the site:** The site is cleaned from rubbles, grass, waste materials, etc.
2. **Mark the layout:** The boundaries of the foundation are marked on the ground with accurate measurements taken from the approved house design drawings/plans
3. **Excavate the foundations:** The soil within the marked boundaries are excavated to the required depth mentioned in the design drawings/plans
4. **Foundation work:** The excavated surface is compacted, leveled and PCC work is done. Then reinforcement bars are tied together and placed in the PCC. For RCC, the framework/shutter for concreting work is placed as per the specifications in the drawings and concrete is poured and packed. For masonry constructions, brick/stone walls are constructed in specified thickness.
5. **Damp proof course (DPC):** At the plinth level (ground floor level of the house, slightly above the ground level), a plinth band is constructed using specified thickness of RCC or timber/bamboo.
6. **Ground floor superstructure:** The structure above the foundation is called superstructure. For RCC, columns/pillars are casted, walls of brick/hollow concrete block/etc. are constructed. A sill band at the base of the window opening and a lintel band just above the doors/windows opening are constructed from RCC. For load bearing masonry, thick walls of specified dimensions are raised and a lintel band of RCC/wood is constructed.
7. **First floor slab:** After the superstructure, beams and slabs are constructed either of RCC or wood. If there are extra storeys, the process is the same, superstructure and slab/roof.
8. **Plaster/Cladding works:** The walls are then plastered with finishes like Cement/Mud mortar or finishes of your choice. There are other options of wall cladding like tiles, aluminum, prefabs, etc. It is your choice to plaster/clad or not to.
9. **Doors and Windows:** Doors and windows are installed.
10. **Fixing electrical and plumbing works:** Necessary electrical and plumbing works are carried out.
11. **Flooring:** Final floor finishing is carried out with preferred finishes like cement punning, tiles, marbles, timber, etc based on the preferred choice.
12. **Painting/Cladding works:** For the plastered walls, you will either need to paint your walls or clad the walls with tiles/marbles.

Implementation: Health and Safety

How to manage my construction?

Once you have identified the personnels for construction, suppliers to supply quality materials and have obtained the temporary building permit from the municipality, you can now start the implementation, i.e. construction of your house. And to begin with, you will need to ensure a safe construction environment for the construction laborers. The minimum safety measures include:

- Provide Personal Protective Equipment (PPE) to all the laborers: You as a homeowner, are responsible to provide minimum PPE like i) Visibility Jacket, ii) Boots, iii) Gloves and iv) Helmet to all laborers, both skilled and unskilled.
- First Aid: At the construction site, you will need to ensure the provision of a first aid kit and also ensure the accessibility of each laborers to the kit.
- Installing signages and barricades: The construction site should be properly barricaded with ropes or other easily accessible barricades so as not to disturb any passers-by and to prevent the entry of external persons. This will help to prevent accidents.



Implementation: Cash Flow Strategy

How can I manage my budget and expenses during the construction period?

To begin the construction, it is very essential to plan and prepare the cash flow strategy, i.e. when and how to manage the fund/cash for the construction expenses. This financial management plan will help you to identify when the expenses are occurring, how much is the expected expense and how to fund for the expense.

The table here provides a brief summary to prepare your cash flow strategy. Suppose you are constructing a two and a half storey RCC house of 1,800 sq.ft. with a tentative cost per square feet of NPR. 3,400, costing a total of NPR. 6,120,000. Further, you also are eligible for

Project Expenditures NPR			Source of expenses & Management NPR			
Phase of construction	Expenses estimate in %	Expense's estimated amount	Tranche	Savings	Monthly Income	Loan
Preparations	5%	40,000	50,000		15,000	
Foundations	20%	160,000		1,000,000	30,000	2,000,000
First Floor	15%	120,000	150,000		30,000	
Second Floor	15%	120,000			30,000	2,000,000
Finishes	45%	360,000	100,000		30,000	2,00,0000
	Total	800,000				

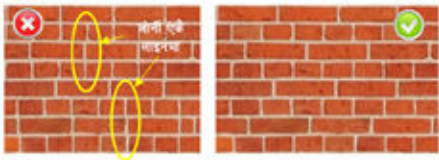
NDRRMA's tranche as landslide-affected. For the expenditure, the tentative percentage for costs for the various phases of construction is similar for every type of construction. You can also consult your contractor or engineer for the cost breakdown based on the stages of construction. Now after the breakdown, you will have a tentative idea of the cost you will require to manage for the various stages of construction. You need to be mindful of the stages when you will be able to get the government tranches and loans, please refer to the Budget section for the details.

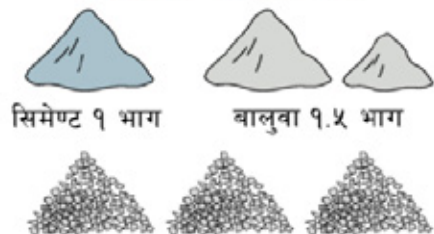
For the Preparations stage, the estimated expense is 5% of the total cost of construction, i.e. NPRs. 306,000. You will be able to access the first tranche of government because you are eligible as a landslide-affected. You are not eligible for any loans from banks at this stage. So, the expense can be managed from the savings and monthly income. After the construction of the foundations, you can apply for bank loans, so you will need to prepare beforehand and complete the documentation process required.

Implementation: Monitoring Workmanship

How to manage my construction/ensure quality workmanship?

Finally, you can now start the construction of your house. You need to monitor this stage very carefully to ensure a safe and compliant house is being constructed. You can get an engineer's supervision for the construction or the contractors too can help you monitor the workmanship of the construction. Some basic construction qualities you will need to monitor and ensure are listed in the table below.

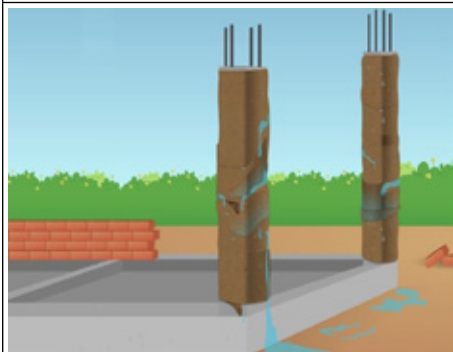
Masonry: Stone or brick walls above ground level	
 ढुंटाको जोडीहत एकै माइनामा परेका ढुंटाको जोडीहत एकै माइनामा नपरेका	Brick Masonry • Bricks should be soaked for at least 7 hours before using for construction. • For a half brick (4-inch thick) wall, cement sand ratio should be at least 1:4, i.e. one part cement and four parts sand, while for full brick (9-inch thick) wall and wall more than 9-inch thick, it should be 1:6, one part cement and six part sand. • Mortar thickness, which is the mortar filling space between bricks in both horizontal and vertical direction, should not be greater than 12 mm. • In one day, a brick wall should be constructed to a maximum height of 3 feet, the construction should not be more than 3 feet. • Horizontal level should be checked in all layers using water level and thread whereas vertical level should be checked using a plumb. • Brick joints should not fall on the same vertical line. If we look into any two adjacent rows of brick, the mortar joint should not be in the same vertical line as shown in the picture beside.
	Stone Masonry • Mortar thickness, which is the mortar filling space between bricks in both horizontal and vertical direction, should not be less than 10mm and more than 20mm • Stone joints should not fall on the same vertical line. If we look into any two adjacent rows of stone, the mortar joint should not be in the same vertical line as shown in the picture beside. • Through stones, i.e., stone equal to the full thickness of wall or not less than three fourth of the thickness of wall should be placed every 2 feet distance vertically and every 4 feet distance horizontally. If through stones are not available, steel "S" hooks can be used. • Corners should be constructed in steps format so that two walls can be connected properly. • In one day, a stone wall should be constructed to a maximum height of 3 feet, the construction should not be more than 3 feet.

Steel reinforcement: Rebars  चित्र १६: डण्डी जोडाने तरिका	- The number and size of steel reinforcements are mentioned in the structural drawings. The minimum number of steel bars for a pillar/column is 8. 4 bars should be of 12mm and 4 bars of 16mm. For a beam, a minimum of 4 bars of 12 mm should be used. - You should not connect two bars at the ends. Whenever the length of the bar needs to be added, ensure good lapping, i.e. overlapping two rebars to a distance equal to 60 times the diameter of the bar. So, for a 16mm bar, the lap should be 960mm (16X60). - You need to ensure minimum cover i.e. concrete layer over all reinforcements. For RCC pillars/columns it is 40mm, for beam it is 25mm, for slab it is 15mm and for foundations it is 50mm.
Concreting Process  Batching	- There are fixed ratios of construction materials you will need to use for preparing concrete and mortars. To get the ratios correct, the materials are batched i.e. measured using a batching box. - Batching boxes generally should be of size 12 inch x 12 inch x 12 inch.
ढलानको मसलाको लागि  Proportions for concreting	- Mixing of cement, sand, aggregate and water in fixed quantities is known as concreting. - There are different grades of concrete used for construction. Grades represent the strength of concrete. Common grades used for different types of construction are M15(1:2:3), M20 (1:1.5:3), M25(1:1:2), etc. M15 means one part cement, two parts sand and three part aggregate. - For construction of houses generally M20 grade concrete is used which is prepared from one part cement, one and a half part sand and three parts aggregate unless otherwise specified in the structural drawings.



Process of preparing concrete mix

- After batching to the right proportion, the first step of preparing the concrete mix is dry mixing of cement and sand. Cement and sand must be thoroughly mixed for at least three times until a uniform color of mix is obtained.
- The second step is adding aggregates in the cement sand mixture and further mixing the mix for at least three times until the mix is uniformly graded.
- The third step is making a crater/hole at the center of the mixture pile to pour in water.
- The final step is adding water in the center of the crater /hole and mixing slowly. This whole process of mixing can be done either manually or using a rotating machine, the process remains the same. Generally, for 1 bag of cement, 20 liters of cement is required.
- Thus prepared wet concrete mix should be used within half an hour and when placing the concrete mix for casting of columns/beams/slab, the mix should be dropped from a height less than three feet.



Curing

- After casting/placing the concrete, concrete starts to set. After six hours of setting, curing should be done. Curing is the process of keeping concrete moist.
- It can be done by spraying water 2-3 times a day, ponding or covering with wet jute bags.
- Curing should be continued for 21 days.



Formwork

- For vertical sides of columns, walls and beams, formwork should be kept for a minimum of 16-24 hours.
- For horizontal sides/soffit of slabs and beams, formwork should be kept for 3 days and 7 days respectively.
- For props/supports to slabs spanning up to 4.5 m and over 4.5m, formwork should be kept for 7 days and 14 days respectively.
- For props/supports to beams and arches spanning up to 6m and over 6m, formwork should be kept for 14 days and 21 days respectively.

Annex: Financial Checklist

SN	Checklist	Yes	No	Remarks
1	NDRRMA, Heritage, Revolving fund			
	Is your name in NDRRMA, Heritage, revolving fund beneficiary list	☐ Yes	☐ No	
	Red Card from Palika/Ward office	☐ Yes	☐ No	
	First tranche from Bank (Reconstruction)	☐ Yes	☐ No	
	DPC level	☐ Yes	☐ No	
	2nd tranche (Reconstruction)	☐ Yes	☐ No	
	Roof level	☐ Yes	☐ No	
	3rd tranche (Reconstruction)	☐ Yes	☐ No	
2	Assets, Income and Savings			
	Do you have a list of assets or property?	☐ Yes	☐ No	
	Valuation of assets or property	☐ Yes	☐ No	
3	Loan			
	Bank account	☐ Yes	☐ No	
	Know your interest rate	☐ Yes	☐ No	
	Loan tenure	☐ Yes	☐ No	
	Citizenship	☐ Yes	☐ No	
	PP size photographs	☐ Yes	☐ No	
	Copy of Telephone bill or bill payment of drinking water	☐ Yes	☐ No	
	Loan Application form	☐ Yes	☐ No	
4	Identification			
	Photograph (2 copies each)	☐ Yes	☐ No	
	Citizenship and/or Passport	☐ Yes	☐ No	
	Relationship Certificate (if applicable)	☐ Yes	☐ No	
	Marriage Certificate (if applicable)	☐ Yes	☐ No	
	Bank (deposit & loan) account statement of at least 6 months	☐ Yes	☐ No	
	Firm/Company Registration Certificate (if applicable)	☐ Yes	☐ No	

SN	Checklist	Yes	No	Remarks
	PAN/VAT Certificate (If applicable)	☐ Yes	☐ No	
	Article of Association (AOA) and Memorandum of Association (MOA) (if applicable)	☐ Yes	☐ No	
	Partnership Deed (if applicable)	☐ Yes	☐ No	
	PAN Certificate (Loan amount equal or above 5 million)	☐ Yes	☐ No	
	PAN Certificate of the individual (for Loan amount above NPR 5 million)	☐ Yes	☐ No	
	Location Map (Residence & Office)	☐ Yes	☐ No	
5	Collateral			
	Land Ownership Certificate (Lalpurja)	☐ Yes	☐ No	
	Rajinama or Bakaspatra or Ansha-banda	☐ Yes	☐ No	
	Malpot Receipt	☐ Yes	☐ No	
	Tax clearance of building	☐ Yes	☐ No	
	Blueprint and trace map	☐ Yes	☐ No	
	Char Killa Pramanit	☐ Yes	☐ No	
	Approved house plan	☐ Yes	☐ No	
	Nirman Sampanna Patra/Abhilekhikaran	☐ Yes	☐ No	
	Ghar Kayam in Lalpurja in case of existence of house	☐ Yes	☐ No	
6	Income Source			
	Salary certificate	☐ Yes	☐ No	
	Rental agreement (If any) along with tenant's identification	☐ Yes	☐ No	
	Audited financials of last 2 fiscal years (If business income)	☐ Yes	☐ No	
	Pension Certificate (Pension Patta)	☐ Yes	☐ No	
	Copy of renewed/valid blue book of vehicle along with route permit (If income is from commercial vehicle)	☐ Yes	☐ No	
	Salary certificate, work permit, valid passport with visa, pay slip and bank account statement (If income is from foreign employment)	☐ Yes	☐ No	
	Latest tax paid receipt or tax clearance certificate for all income (wherever applicable)	☐ Yes	☐ No	

Annex: Technical Checklist

S.N	Category	Checklist	Yes or No		Remarks
1	Site Selection	Geological fault or ruptured area	☐ Yes	☐ No	
		Steep Slope > 20°	☐ Yes	☐ No	
		Landslide susceptible Area	☐ Yes	☐ No	
		Riverbank and Waterlogged Area	☐ Yes	☐ No	
		Rock-fall Area	☐ Yes	☐ No	
		Filled Area	☐ Yes	☐ No	
2	Shape and Size of house	Is the shape of the house regular, square or rectangle?	☐ Yes	☐ No	
		Are there at least two number of bays?	☐ Yes	☐ No	
		Is the length less than or equal to 3 times breadth?	☐ Yes	☐ No	
		Is the area of a room or between four pillars less than 13.5 sq m ?	☐ Yes	☐ No	
3	Layout	Is proficient technical person involved during layout	☐ Yes	☐ No	
		Are the stakes set out permanently till the construction of plinth level?	☐ Yes	☐ No	
		Is the benchmark set for plinth level?	☐ Yes	☐ No	
		Is the foundation depth measured from the plinth level benchmark?	☐ Yes	☐ No	
4	Construction Workmanship	Construction material quality			
		Check specifications before buying construction materials?	☐ Yes	☐ No	
		Check the quality of construction materials?	☐ Yes	☐ No	
		Are the materials stored properly near the construction site?	☐ Yes	☐ No	
		Masonry			
		Over burnt, under burnt and deformed brick is not used.	☐ Yes	☐ No	
		Water immersion for 7 hours	☐ Yes	☐ No	
		Joints not in same vertical line	☐ Yes	☐ No	
		For half brick wall mortar ratio 1:4 and for full brick wall mortar ratio 1:6	☐ Yes	☐ No	
		Mortar thickness 10mm	☐ Yes	☐ No	
		Curing for 7 days	☐ Yes	☐ No	

		Concrete			
		Foundation, columns, beam, and slab concrete grade at least M20	☐ Yes	☐ No	
		Concrete mixing surface clean	☐ Yes	☐ No	
		Use of batching box	☐ Yes	☐ No	
		Cement used within 3 months of manufacture date	☐ Yes	☐ No	
		Aggregate well -graded	☐ Yes	☐ No	
		Used mixed for concreting foundation, beam, pillar and slab	☐ Yes	☐ No	
		Use vibrator while concreting	☐ Yes	☐ No	
		Use of wet concrete within half an hour	☐ Yes	☐ No	
		Use of cover block	☐ Yes	☐ No	
		Cover in foundation 2 inch, pillar 1.5 inch, beam 1 inch and slab 0.5 inch-	☐ Yes	☐ No	
		Water quantity as required	☐ Yes	☐ No	
		Curing for 21 days	☐ Yes	☐ No	
		Rebar			
		Cutting, bending, placing in presence of technical person	☐ Yes	☐ No	
		Use of chair for separating rebars	☐ Yes	☐ No	
		Not more than 50% overlapping in one place	☐ Yes	☐ No	
		Pillar joint in the middle and at least 60 times dia overlap	☐ Yes	☐ No	
		Formwork			
		Proper placement of formwork before concreting	☐ Yes	☐ No	
		Formwork removing date mentioned	☐ Yes	☐ No	
		Use of formwork in foundation	☐ Yes	☐ No	

The detailed checklist for various stages of construction for different construction typology can be accessed through:

https://www.hrrpnepal.org/uploads/media/TxFAPwktdQmuvXeoEas1_2018_08_13.pdf

